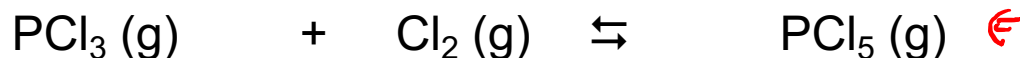


Learning Objectives

1. Calculate equilibrium constant given concentrations
2. Calculate equilibrium concentrations given K_c

Calculating K from Measured Equilibrium Concentrations

0.100 mol PCl_3 and 0.0500 mol Cl_2 are added to a 1.00 L sealed container. After reaction came to equilibrium at 25 °C, flask contained 0.0600 mol $\text{PCl}_3(\text{g})$. What is the equilibrium constant for this reaction at 25 °C?



Initial concentration (M)	0.100	0.0500	0
concentration Change during the reaction (M)	-0.0400	$-0.0400 \text{ PCl}_3 \times \frac{1 \text{ Cl}_2}{1 \text{ PCl}_3}$	$+0.0400 \text{ PCl}_3 \times \frac{1 \text{ PCl}_5}{1 \text{ PCl}_3}$
Final concentration (M)	0.0600	0.0100	0.0400

The change of concentration follows stoichiometry!

$$K = \frac{[\text{PCl}_5]_e}{[\text{PCl}_3]_e \cdot [\text{Cl}_2]_e} = \frac{(0.0400 \text{ M})}{(0.0600 \text{ M}) \times (0.0100 \text{ M})} = 66.7$$

Your turn!

The following equilibrium has been studied at 780 °C:



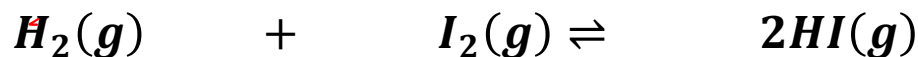
Initially the reaction mixture contains $[\text{CO}] = 0.500\text{M}$ and $[\text{H}_2] = 1.00\text{M}$. At the equilibrium, $[\text{CO}]$ is found to be 0.150M and the volume of the mixture is not changed. Calculate the equilibrium constant (K_c) of the reaction at this temperature.



Initial concentration (M)	0.500	1.00	0
concentration Change during the reaction (M)			
Final concentration (M)	0.150		

Calculating Equilibrium Concentrations from K

A mixture of 0.500 mol H_2 and 0.500 mol I_2 was placed in a 1.00-L sealed flask at 430°C . The equilibrium constant, K_c , for the reaction is 54.3 at this temperature. Calculate the concentrations of H_2 , I_2 , and HI at equilibrium.



Initial concentration (M)	0.500	0.500	0
concentration Change during the reaction (M)	-x	-x	+2x
Final concentration (M)	<u>0.500 - x</u> 0.167	<u>0.500 - x</u> 0.167	<u>2x</u> 0.786

The change of concentration follows stoichiometry!

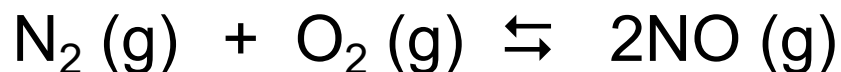
$$K = 54.3 = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(2x)^2}{(0.500-x)(0.500-x)} = 54.3$$

$$\frac{2x}{0.500-x} = 7.37 \rightarrow 2x = 3.68 - 7.37x$$

$$9.37x = 3.68 \quad x = 0.393$$

Your turn!

Consider the reaction below. A reaction has a $K_c = 0.1$ at $2000\text{ }^\circ\text{C}$. A reaction mixture at $2000\text{ }^\circ\text{C}$ initially contains 0.200 M N_2 and 0.200 M O_2 . What will be the equilibrium concentration of N_2 , O_2 and NO in the container?

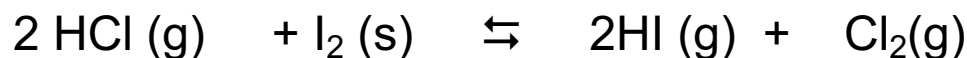


Initial concentration (M)			
concentration Change during the reaction (M)			
Final concentration (M)			

0.173 M; 0.173 M; 0.054 M

Your turn!

A reaction has a $K_c = 1.6 \times 10^{-34}$ at 25°C . Suppose 0.400 mol HCl and 0.400 mol I_2 are placed in a 4.00L container. What will be the equilibrium concentration of HI and Cl_2 in the container?



Initial concentration (M)	0.100		0	0
concentration Change during the reaction (M)	$-2X$		$+2X$	$+X$
Final concentration (M)	$0.1 - 2X$ 0.1		$2X$ 1.47×10^{-12}	X 7.37×10^{-13}

If the K is very very small ($K < 10^{-3}$), the usage of reactants can be ignored.

If the K is very very large ($K > 10^3$), forward reaction goes close to completion and limiting reactant is used up.

$$K_c = \frac{[\text{HI}]^2 [\text{Cl}_2]}{[\text{HCl}]^2} = \frac{(2X)^2 \cdot (X)}{(0.1 - 2X)^2} = 1.6 \times 10^{-34}$$

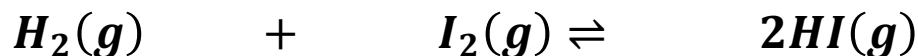
$$0.1 - 2X \approx 0.1$$

$$\frac{4X^3}{0.1^2} = 1.6 \times 10^{-34}$$

$$X = 7.37 \times 10^{-13}$$

Calculating Equilibrium Concentrations from K

A mixture of **0.00623 M** H_2 , **0.00414 M** I_2 and **0.0224 M** HI was placed in a sealed flask at 430 °C. The equilibrium constant, K_c , for the reaction is **54.3** at this temperature. Calculate the concentrations of H_2 , I_2 , and HI at equilibrium.



Initial concentration (M)			
concentration C hange during the reaction (M)			
Final concentration (M)			

The **change** of concentration follows stoichiometry!



$$50.3x^2 - 0.654x + 8.98 \times 10^{-4} = 0$$

This is a quadratic equation of the form $ax^2 + bx + c = 0$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Announcement 4/22

Assignments that are due soon

- Pre-Class Assignment Week 12-F

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

1st Chem RePS or SI worksheets for Unit 4 will be closed on 4/29

Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom

SI Session

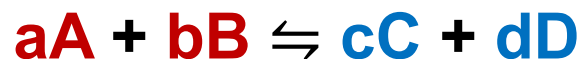
Saige Douglass – Wed 5:30 – 7:30 PM at Hashinger 412

Paige Kwan - Thu 5 – 6 PM at KC134

Learning Objectives

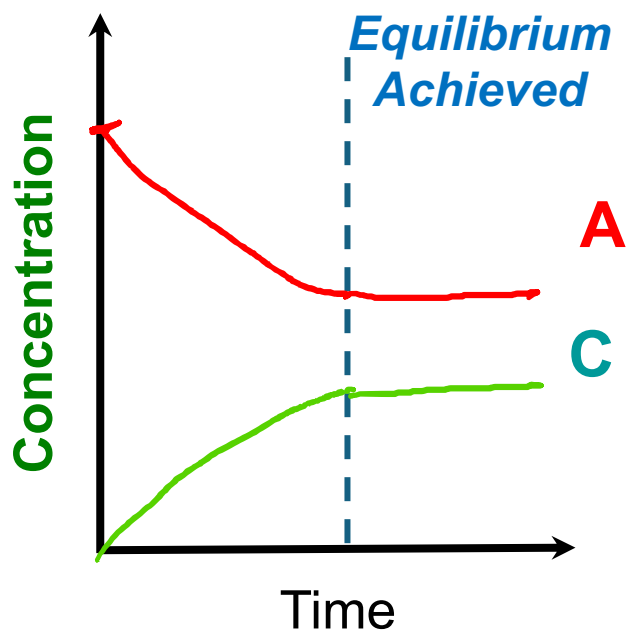
1. Predict the direction of change based on the comparison of Reaction Quotient, Q and Equilibrium Constant, K_c
2. Determine the effect of a disturbance on Equilibrium, applying L'Chatelier's Principle
 - Concentration Change
 - Volume Change
 - Pressure Change
 - Temperature Change
 - Addition of Catalyst

Review on Dynamic Equilibrium



$$K = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Reach over time



0 reactant backward favored 1 product forward favored ∞

Dynamically

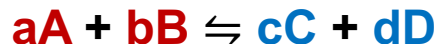
Macroscopically: **no change**

(concentration of reactants and products *remain* constant)

Microscopically: **change continuously**

(Forward and reverse reactions are occurring *indefinitely* and *at exactly the* same rate)

Reaction Quotient (Q)



$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Equilibrium constant (K) uses concentration at equilibrium.

Reaction Quotient (Q) uses concentration at any time.

- $Q_c = 0$ is when the reaction only contains reactant.
- $Q_c < K_c$: the reaction proceeds forward to reach equilibrium
- $Q_c = K_c$: the reaction reaches to equilibrium
- $Q_c > K_c$: the reaction proceeds backward to reach equilibrium
- $Q_c = \infty$ is when the reaction only contains product.



Reactants $\rightarrow$ Products



Equilibrium : no net change



Reactants $\leftarrow$ Products

Your turn!

K_c for the formation of HI from H_2 and I_2 in the gas phase:

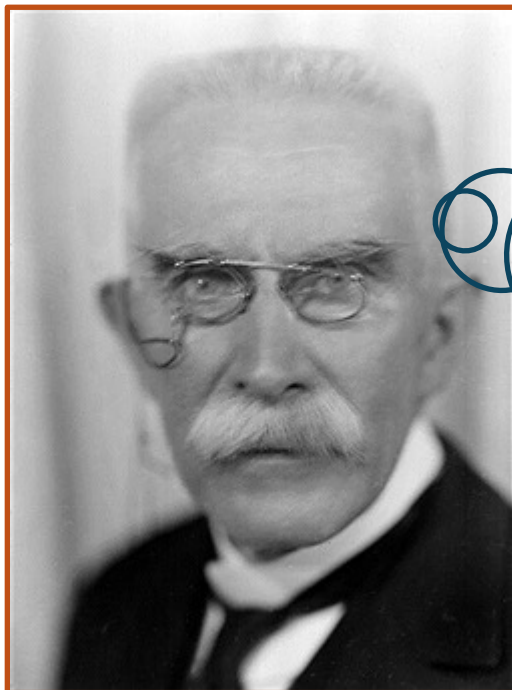


is 54.3 at 430 °C. Suppose that we place **0.243 mol** of H_2 , **0.146 mol** of I_2 , and **1.98 mol** of HI in a **2.00-L** container at 430°C. In which direction will the reaction proceed?

$Q =$

$>$
 $= K$
 $<$

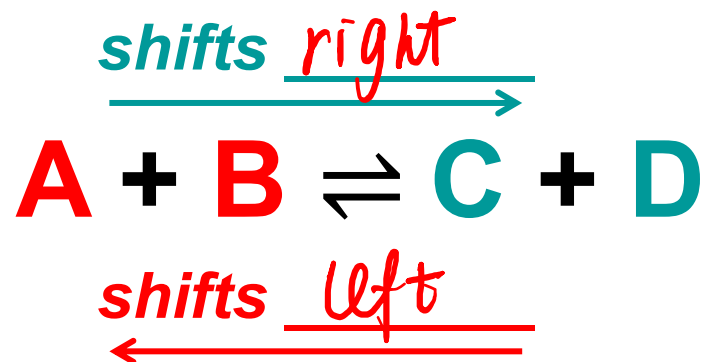
Le Châtelier's Principle



<http://www.gettyimages.com/detail/news-photo/henry-louis-le-chatelier-chemist-and-french-metalworker-news-photo/55756057#henry-louis-le-chatelier-chemist-and-french-metalworker-france-by-picture-id55756057>

If a change
(concentration / volume / pressure
of reacting species, or
temperature) is imposed on a
system at equilibrium, the
position of equilibrium will **shift**
in a direction that tends to
reduce the disturbance.

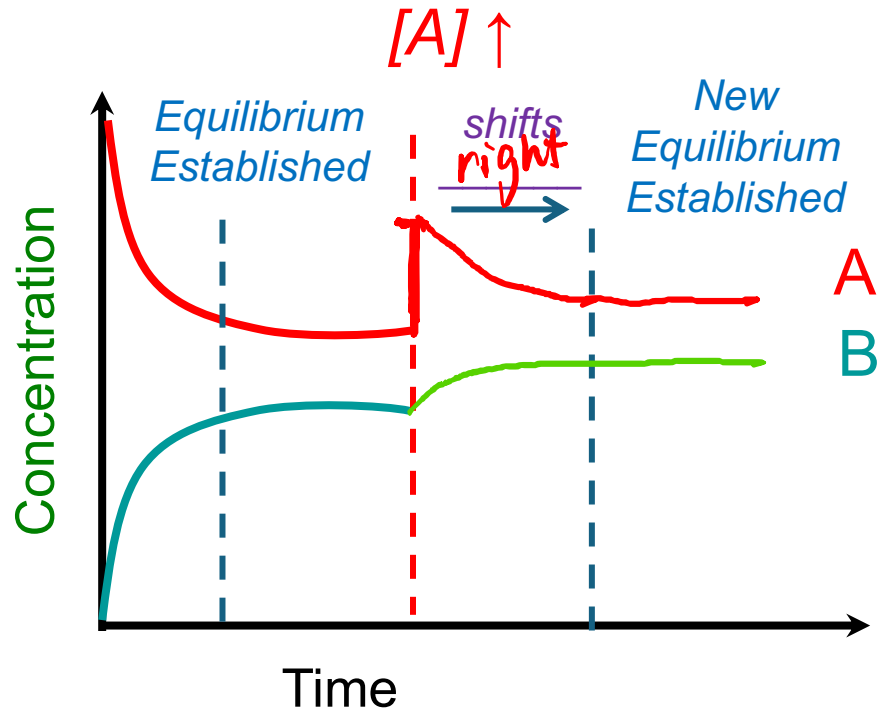
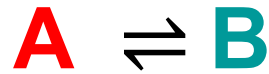
Shifting Equilibrium



When an equilibrium changes such that it favors the production of products (C and D) and the reaction moves to forward direction, we say the equilibrium shifts right

When an equilibrium changes such that it favors the production of reactants (A and B) and the reaction moves to backward direction, we say the equilibrium shifts left

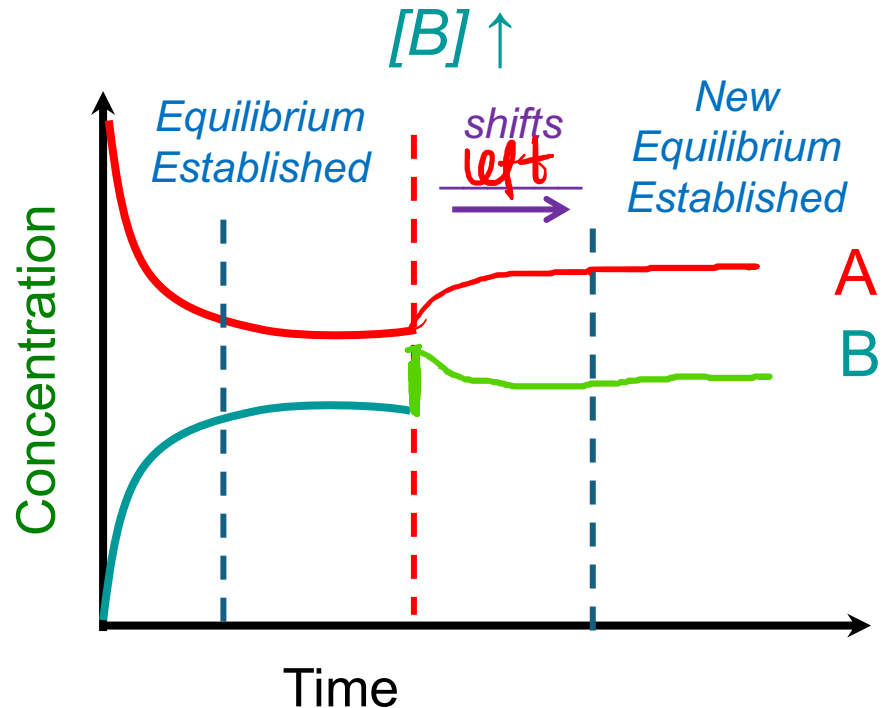
Le Châtelier's Principle: Changing concentration



- Increase [reactants] after equilibrium:

$$Q = \frac{[B]}{[A]} \uparrow < K$$

The equilibrium shift to right to reduce the change



- Increase [products] after equilibrium:

$$Q = \frac{[B]}{[A]} \uparrow > K$$

The equilibrium shift to left to reduce the change

Case Study



Add NaSCN to
increase $[\text{SCN}^{-}]$

Equilibrium shifts to left;
Color becomes more red

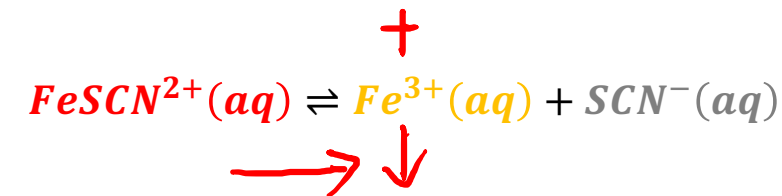


Initial solution at equilibrium

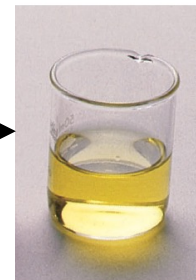
Add $\text{Fe}(\text{NO}_3)_3$ to
increase $[\text{Fe}^{3+}]$



Equilibrium shifts to left;
Color becomes more red



Add $\text{H}_2\text{C}_2\text{O}_4$
to decrease $[\text{Fe}^{3+}]$

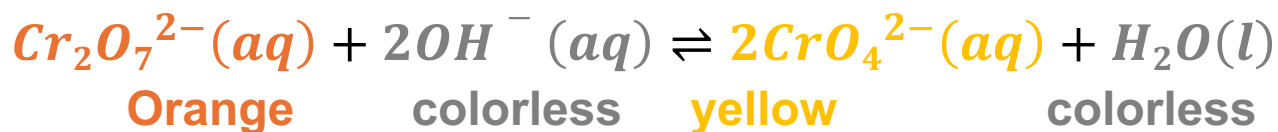


Equilibrium shifts to right;
Color becomes more yellow



+

Your turn!



The equilibrium will shift to ...

1) Adding NaCl (s) $\rightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
no change

2) Adding NaOH (s) $\rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$
 $\uparrow \rightarrow$ right

3) Adding HCl (l) $\rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
 $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$
 $\downarrow \leftarrow$ left



Your turn!

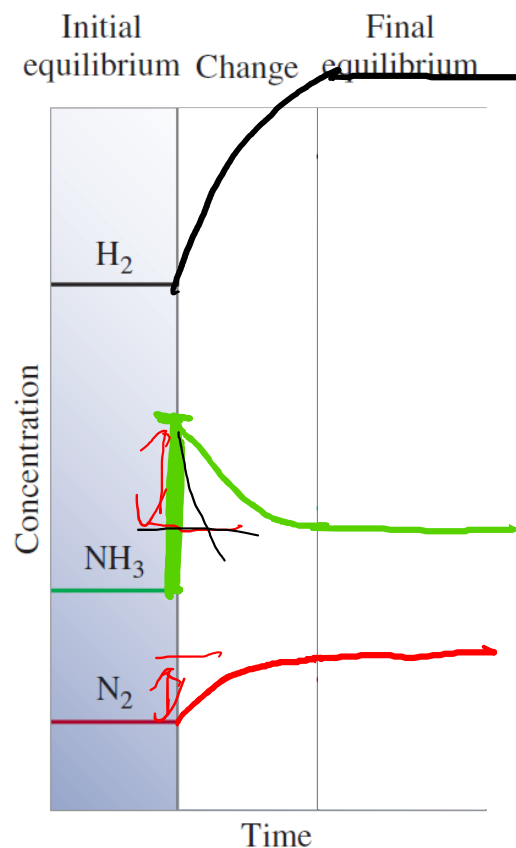
At 720 °C, the equilibrium constant, K_c , for the reaction



is 2.37×10^{-3} . In a certain experiment, the equilibrium concentrations are $[\text{N}_2] = 0.683 \text{ M}$, $[\text{H}_2] = 8.80 \text{ M}$, and $[\text{NH}_3] = 1.05 \text{ M}$. Suppose some more NH_3 is added to the mixture so that its concentration is increased to 3.65 M .

- (a) Use Le Châtelier's principle to predict in which direction will the reaction shift to reach a new equilibrium and draw the shift on the diagram below.

I	0.683	8.80	3.65
C	+x	+3x	-2x
E	<u>0.683+x</u>	<u>8.80+3x</u>	<u>3.65-2x</u>



Your turn!

At 720 °C, the equilibrium constant, K_c , for the reaction



is 2.37×10^{-3} . In a certain experiment, the equilibrium concentrations are $[\text{N}_2] = 0.683 \text{ M}$, $[\text{H}_2] = 8.80 \text{ M}$, and $[\text{NH}_3] = 1.05 \text{ M}$. Suppose some NH_3 is added to the mixture so that its concentration is increased to 3.65 M .

(b) Confirm your prediction by calculating the reaction quotient, Q_c , and comparing its value with K_c .

Le Châtelier's Principle: Changing Volume

$\{ \begin{matrix} (g) \\ (aq) \end{matrix} \}$ add solvent

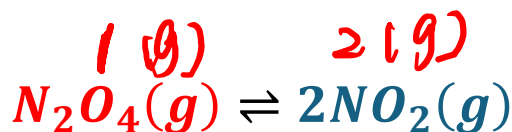
The direction of equilibrium shift depends on the total n number of gaseous / solute molecules

→ more concentrated

$$PV = nRT$$

- Decrease volume: shift to the direction to produce less moles of molecules
- Increase volume: shift to the direction to produce more moles of molecules

→ Dilute



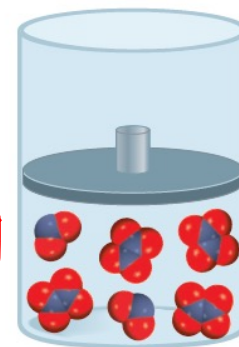
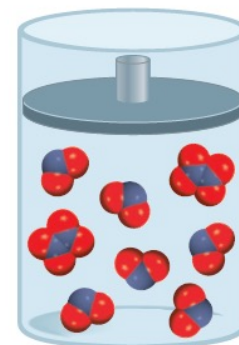
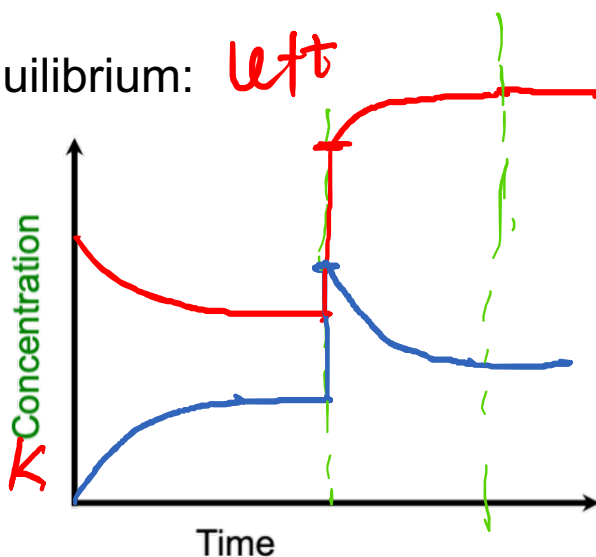
solute (aq) (g)

- Decrease the volume by 1/2 after equilibrium: left

$$[N_2O_4]' = \frac{[N_2O_4]_e}{\frac{1}{2}} = 2[N_2O_4]_e$$

$$[NO_2]' = 2[NO_2]_e$$

$$Q' = \frac{(2[NO_2]_e)^2}{2[N_2O_4]_e} = \frac{4[NO_2]_e^2}{2[N_2O_4]_e} = 2K > K$$



The new equilibrium shift to left to reduce the change (more concentrated)

Announcement 4/24

Assignments that are due soon

- Pre-Class Assignment Week 13-M

Office Hour

- Dr. Wang's: Fri 2-3 PM at KC 382



Sensory Evaluation
TASTE TEST

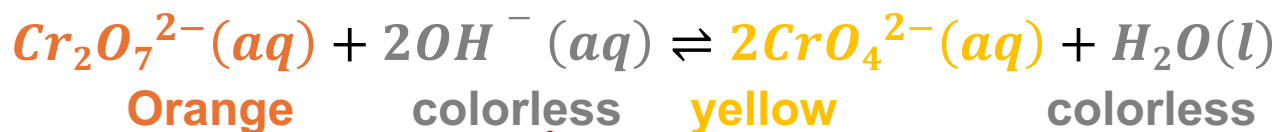
FRIDAY APRIL 24TH

**FREE SNACKS
FOR YOUR
TIME**

KECK 136 | 10:00AM-4:00PM

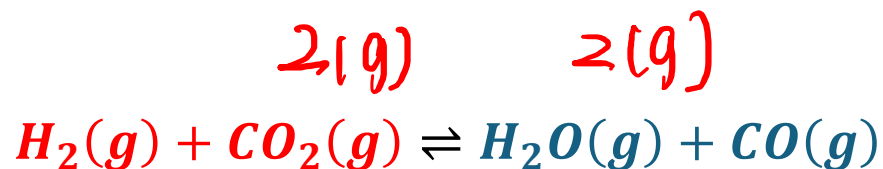
The poster features a teal border and a light beige background. At the top, the title 'Sensory Evaluation' is in large teal font, with 'TASTE TEST' in a smaller, italicized teal font below it. A teal rounded rectangle contains the date 'FRIDAY APRIL 24TH' in white. To the left of a yellow starburst is a QR code. The starburst contains the text 'FREE SNACKS FOR YOUR TIME' in red. At the bottom, a teal bar contains the location and time 'KECK 136 | 10:00AM-4:00PM' in white.

Your turn!



The equilibrium will shift to ... $3(\text{aq}) \leftarrow 2(\text{aq})$

1) Adding more water $\rightarrow V \uparrow \rightarrow$ dilute $\rightarrow$ shift left



The equilibrium will shift to ...

1) Decrease the volume of the reaction mixture $\rightarrow$ concentrated

stay the same

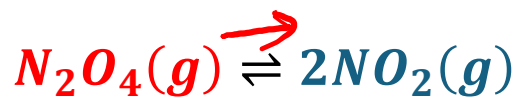


Le Châtelier's Principle: Changing Volume

The direction of equilibrium shift depends on the total number of gas (g) / (aq) solute molecules

$$PV = nRT$$

- Decrease volume: $\rightarrow$ more concentrated shift to the direction to produce less moles of molecules
- Increase volume: $\rightarrow$ more dilute shift to the direction to produce more moles of molecules

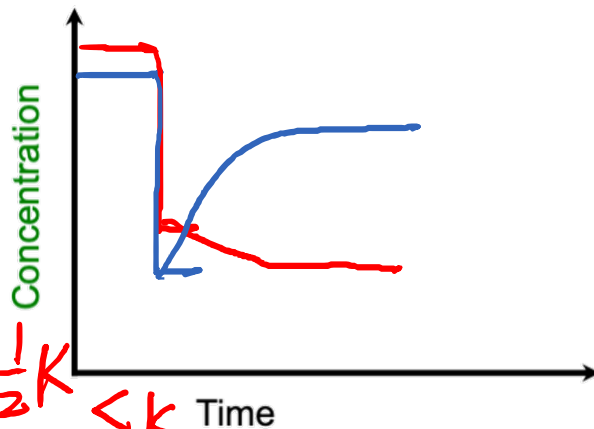


- Increase the volume by 2 after equilibrium: $\rightarrow$ more dilute shift to right

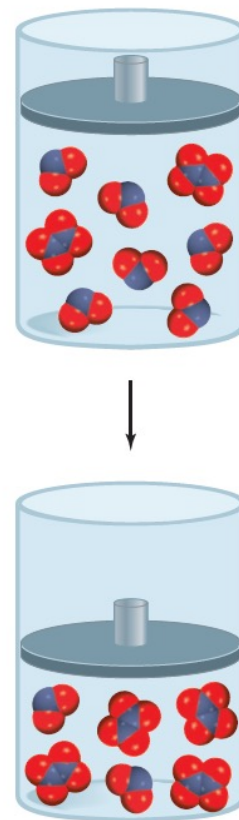
$$[N_2O_4]' = \frac{[N_2O_4]_e}{2}$$

$$[NO_2]' = \frac{[NO_2]_e}{2}$$

$$Q' = \frac{\left(\frac{[NO_2]_e}{2}\right)^2}{\left(\frac{[N_2O_4]_e}{2}\right)} = \frac{\frac{1}{4} [NO_2]_e^2}{\frac{1}{2} [N_2O_4]_e} = \frac{1}{2} K < K$$



The new equilibrium shift to right to reduce the change (diluted)



Le Châtelier's Principle: Changing Pressure

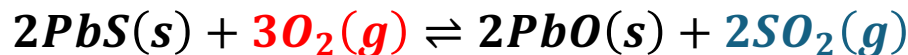
$$PV = nRT$$

Constant *number* of gaseous reactants/products:

- Increase pressure = decrease volume: *more concentrated*
shift to the direction to produce less moles of molecules
- Decrease pressure = increase volume: *more diluted*
shift to the direction to produce more moles of molecules

Constant *volume*:

- Increase pressure by adding an inert gas = fixed volume:
no effect on equilibrium



The equilibrium will shift to ...

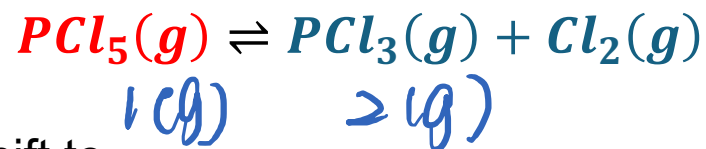
- 1) Increase the pressure and the particles number is not changed:

more concentrated → less molecules → shift right

- 2) Increase the pressure by adding Helium gas into the reaction at fixed volume:

concentration doesn't change → stay the same

Your turn!



The equilibrium will shift to ...

1) Increase the pressure:



less molecules

left

2) Increase the pressure by adding Helium gas into the reaction at fixed volume:



stay the same



Le Châtelier's Principle: Changing Temperature

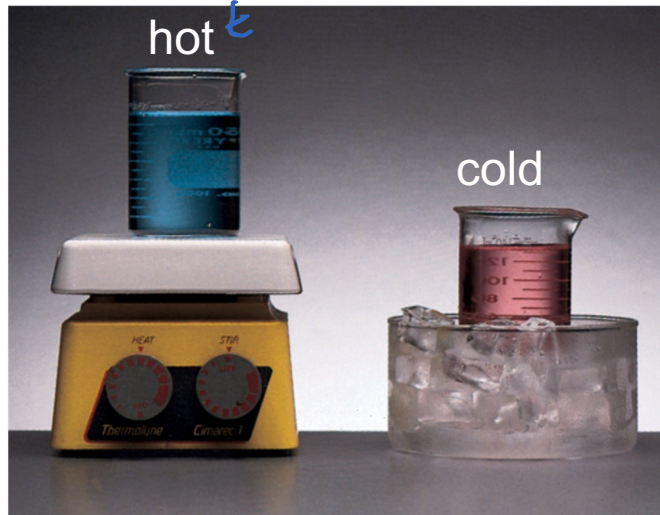
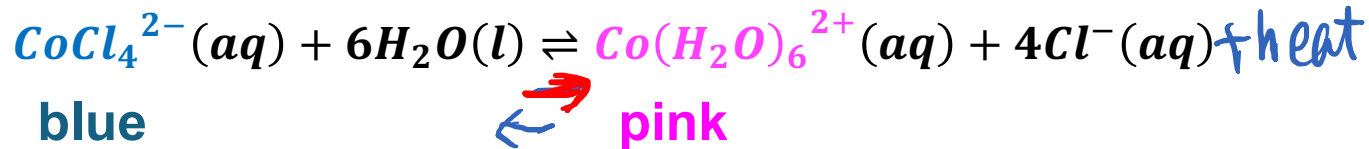
- Increase temperature: shift to the direction to absorb heat ↵
- Decrease temperature: shift to the direction to release heat

For exothermic reaction:



$$\Delta H = (-)$$

- Increase temperature: shift to left
- Decrease temperature: shift to right



Le Châtelier's Principle: Changing Temperature

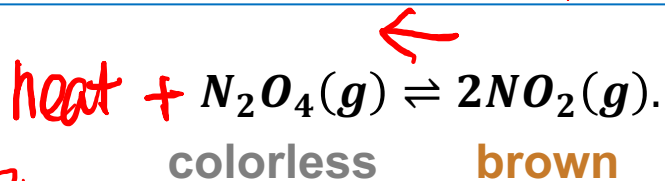
- Increase temperature: shift to the direction to **consume** heat
- Decrease temperature: shift to the direction to **produce** heat

For Endothermic reaction:



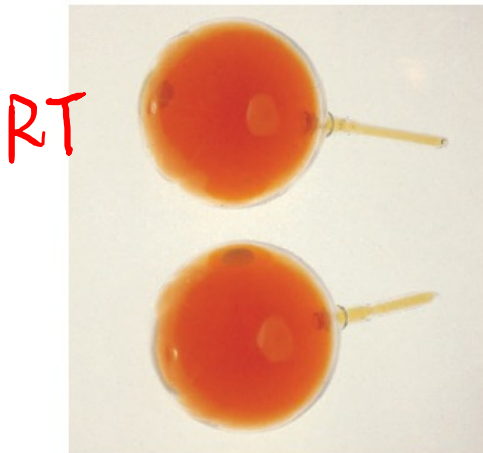
$$\Delta H = (+)$$

- Increase temperature: shift to right
- Decrease temperature: shift to left



$$\Delta H = +58.0 \text{ kJ/mol}$$

Identical to start with

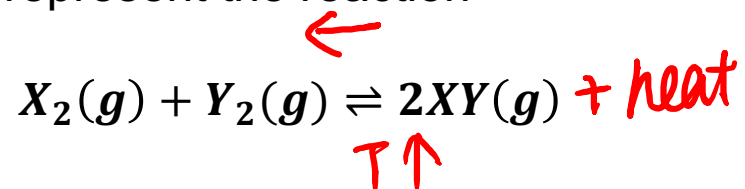


Which beaker is colder?
A. Left B. right

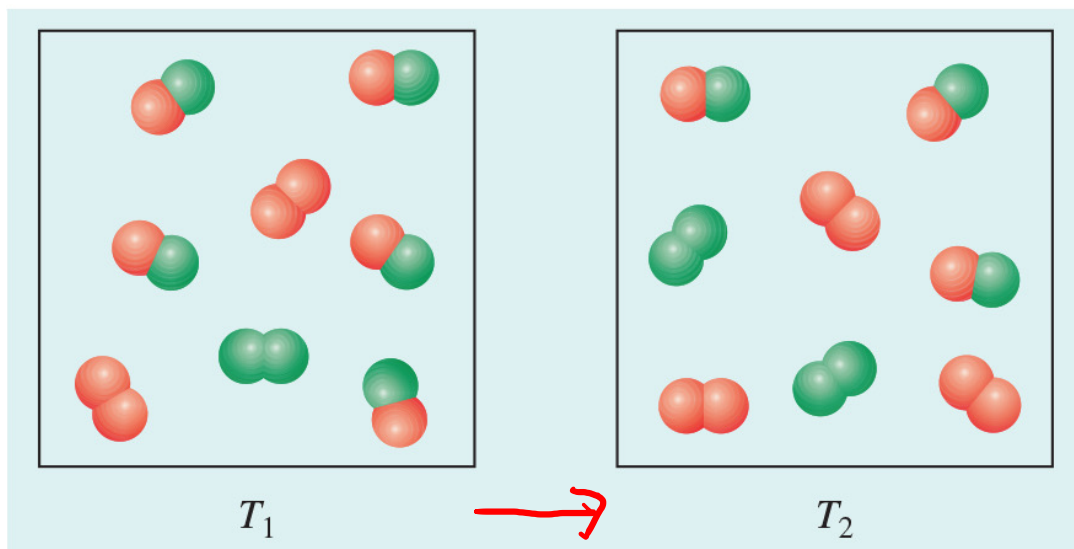


Your turn!

The diagrams shown here represent the reaction



at equilibrium at two temperatures ($T_2 > T_1$). Is the reaction endothermic or exothermic?



More XY

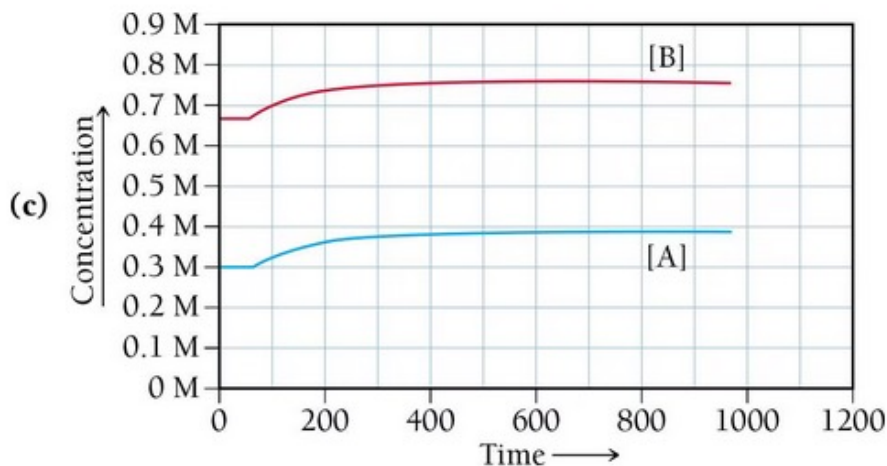
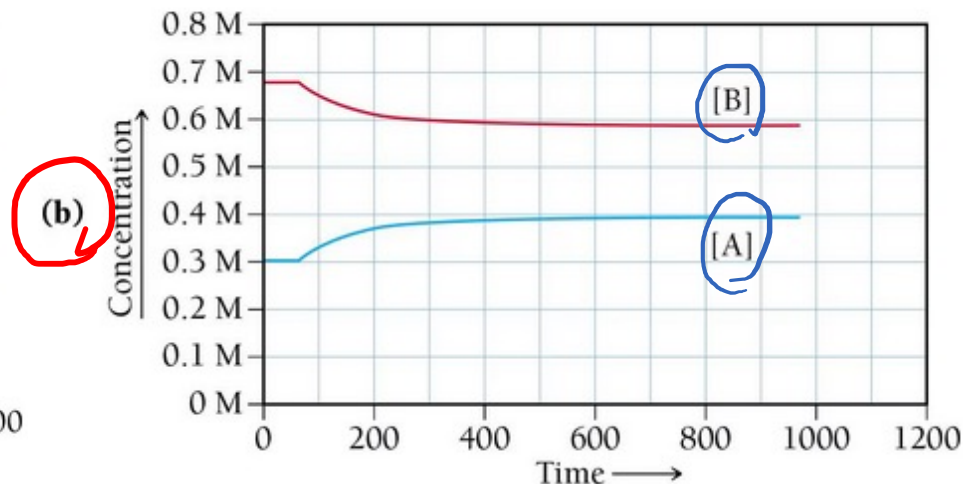
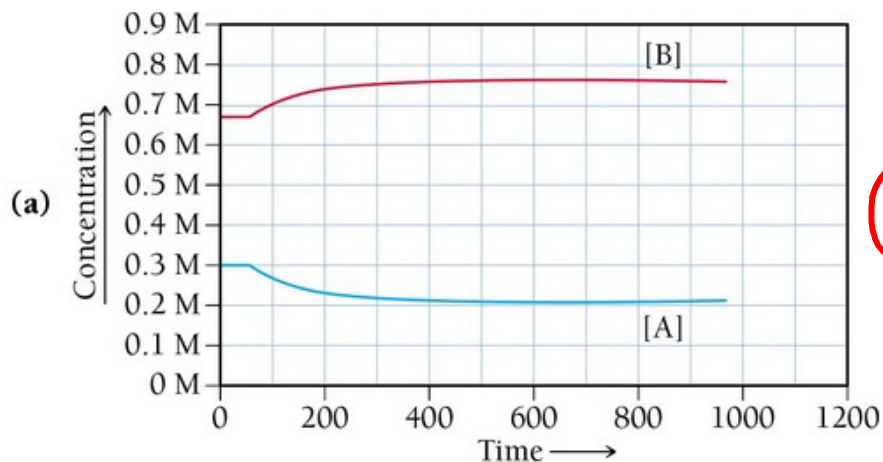
More X_2 and Y_2



Your turn!



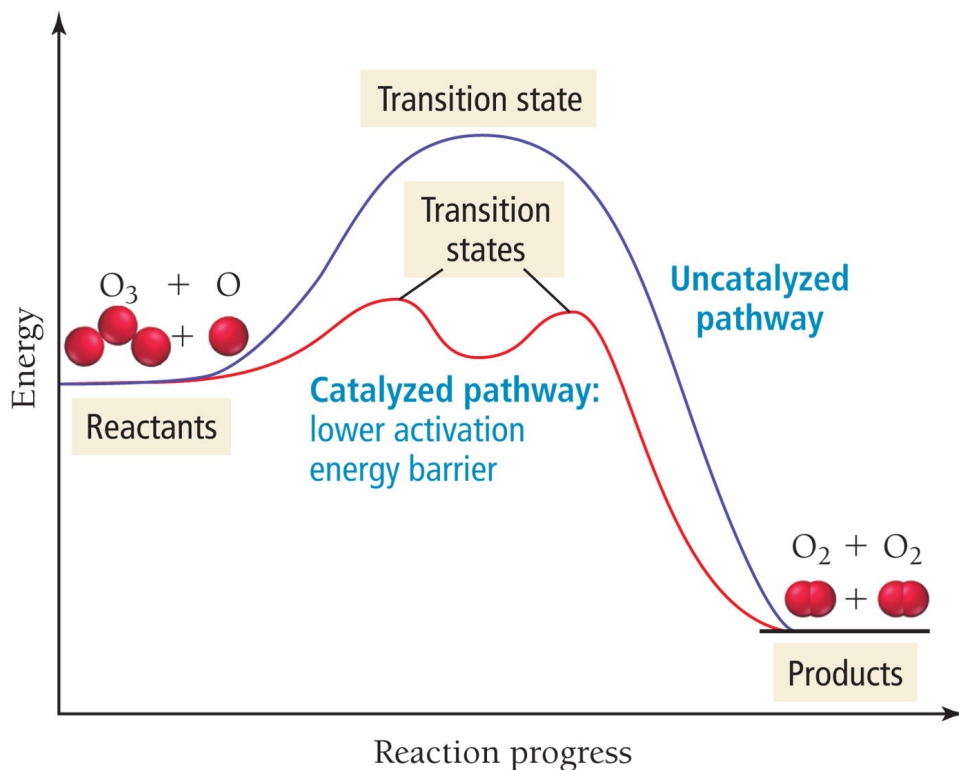
The reaction $A(g) \rightleftharpoons B(g)$ is exothermic. The reaction was at equilibrium and then disturbed by increasing the temperature quickly. The reaction is then allowed to come back to equilibrium at the elevated temperature. Which graph represents the concentrations of A and B during this process?



Le Châtelier's Principle: Adding Catalysts

- Catalysts can change the speed of the reaction by lowering E_a
- Catalysts cannot change the equilibrium of the reaction

Energy Diagram for Catalyzed and Uncatalyzed Pathways



Announcements 4/27

Assignments that are due soon

- Pre-Class Assignment Week 13-W

Office Hour

- Dr. Wang's: Tue 10-11 AM at KC 230
- Paige Kwan's: Mon 11:30 AM -12:30 PM at the TLC (Demille 160P)
- Saige Douglass's: Tue 12:00-1:00 PM at the TLC (Demille 160P)

Summary on Le Châtelier's Principle

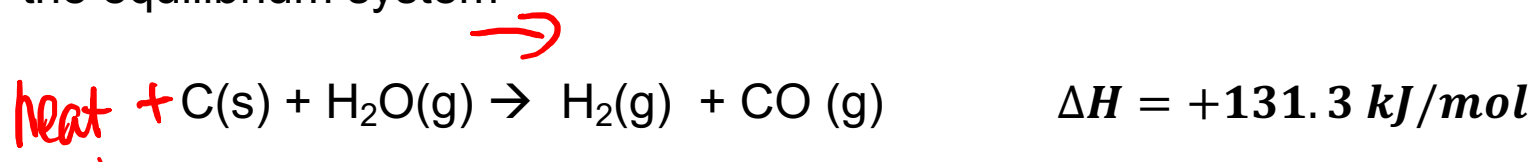


New equilibrium shift to the direction to reduce the introduced changes...

Change	Shift Equilibrium?	If So, How? <i>Q ≠ K</i>	Change K ?
Concentration <i>(g) (aq)</i>	<u>yes</u>	increase [reactant], shift to <u>right</u> decrease [product], shift to <u>right</u> increase [product], shift to <u>left</u> decrease [reactant], shift to <u>left</u>	<u>no</u>
Volume / Pressure	yes, but only when Δn <i>(g) or (aq)</i>	<i>→ more concentration</i> P↑ or V↓, shift to <u>less</u> moles of gases P↑ at fixed V, no effect <i>e</i> <i>→ diluted</i> P↓ or V↑, shift to <u>more</u> moles of gases	<u>no</u>
Temperature	<u>yes</u>	T↑, shift to the direction <u>consume</u> heat T↓, shift to the direction <u>produce</u> heat	<u>yes</u>
Catalysts	<u>no</u>		<u>no</u>

Your turn!

For the equilibrium system



determine in which direction the equilibrium will shift in response to each change:

(a) increase in temperature

$T \uparrow$

consume heat

right

(b) decrease in pressure without adding or remove any gaseous molecules

$\downarrow \uparrow$
 $PV = nRT$

$\rightarrow$ increase $V \rightarrow$ dilute $\rightarrow$ $[1 \rightleftharpoons 2] \rightarrow$ right

(c) addition of $\text{H}_2\text{(g)}$

shift to left

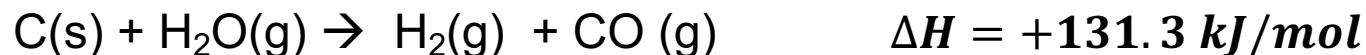
(d) Removing of C(s)

$\uparrow$

no effect

Your turn!

For the equilibrium system



determine in which direction the equilibrium will shift in response to each change:

(a) increase in temperature

(b) decrease in pressure without adding or remove any gaseous molecules

(c) addition of $\text{H}_2(\text{g})$

(d) Removing of C(s)

Learning Objectives

- Relate Gibbs Free Energy and Chemical Equilibrium.

Gibbs Free Energy and Chemical Equilibrium

$$\Delta G = \Delta H - T\Delta S$$

Under constant pressure and temperature:

- $\Delta G < 0$, then the reaction is spontaneous
- $\Delta G = 0$, then the reaction is at equilibrium
- $\Delta G > 0$, then the reaction is nonspontaneous (Backward reaction is spontaneous)

$$\Delta G = \Delta G^\circ + RT \ln Q$$

- At the equilibrium: $Q=K$

$$\Delta G = 0 = \Delta G^\circ + RT \ln K$$

$$\Delta G^\circ = -RT \ln K$$
- ΔG° is when the reaction is at standard condition. (P = 1 atm; [A] = 1 M)

K	ln K	ΔG°	Comments
> 1	<u>> 0</u>	<u>< 0</u>	<u>products</u> are favored over <u>reactants</u> at equilibrium.
= 1	<u>= 0</u>	<u>= 0</u>	Products and reactants are <u>equally</u> favored at equilibrium.
< 1	<u>< 0</u>	<u>> 0</u>	<u>reactants</u> are favored over <u>products</u> at equilibrium.

Your turn! $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -RT \ln K$

A reaction has a positive ΔH° and a negative ΔS° . Is this reaction spontaneous?

- A. Spontaneous
- ☒ B. Nonspontaneous
- C. Depends on temperature

Is the equilibrium constant (K) for this reaction greater than 1, equal to 1, or less than 1?

- A. Greater than 1
- B. Equal to 1
- ☒ C. Less than 1

$\Delta G^\circ > 0$ backward

$$\ln K < 0$$

$$K < 1$$

$$K = \frac{[P]}{[R]}$$



Your turn! $\Delta G = \Delta G^\circ + RT \ln Q$

For the reaction:



$\Delta G^\circ = 1.95 \text{ kJ/mol}$ at 25 °C. What is ΔG (in kJ/mol) at 25 °C when the concentrations are $[\text{A}] = [\text{B}] = 0.315 \text{ M}$ and $[\text{C}] = 0.405 \text{ M}$? What direction will the reaction go? $R = 8.314 \text{ J/(mol K)}$; $T(\text{K}) = T(^{\circ}\text{C}) + 273.15$

$-1.54 \frac{\text{kJ}}{\text{mol}}$; Forward

Your turn! $\Delta G = \Delta G^\circ + RT \ln Q$

For the reaction:



$\Delta G^\circ = 1.95 \text{ kJ/mol}$ at 25°C . What is ΔG (in kJ/mol) at 25°C when the concentrations are $[\text{A}] = [\text{B}] = 0.315 \text{ M}$ and $[\text{C}] = 0.405 \text{ M}$? What direction will the reaction go?

$$\Delta G = 1.95 \frac{\text{kJ}}{\text{mol}} + 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \times (25 + 273.15) \text{ K} \times \frac{1 \text{ kJ}}{10^3 \text{ J}} \times \ln \left(\frac{0.315 \times 0.315}{0.405} \right)$$

$$= 1.95 \frac{\text{kJ}}{\text{mol}} + (-3.49 \frac{\text{kJ}}{\text{mol}})$$

$$= -1.54 \frac{\text{kJ}}{\text{mol}} < 0$$

Forward

$-1.53 \frac{\text{kJ}}{\text{mol}}$; Forward